for the lowering operator, and

$$\mathbf{a}^{+} = (\mathbf{P} + i\omega\mathbf{X})$$

for the raising operator. But history sometimes preempts the obvious. Historically, the raising and lowering operators have been defined with an extra factor in front of them. Here are the official definitions:

$$\mathbf{a}^{-} = \frac{i}{\sqrt{2\omega\hbar}}(\mathbf{P} - i\omega\mathbf{X}), \qquad (10.20)$$

$$\mathbf{a}^{+} = \frac{-i}{\sqrt{2\omega\hbar}}(\mathbf{P} + i\omega\mathbf{X}), \qquad (10.21)$$

If we use these definitions, the Hamiltonian starts to look very simple:

$$\mathbf{H} = \omega\hbar(\mathbf{a}^{+}\mathbf{a}^{-} + 1/2). \qquad (10.22)$$

There are only two properties of $\mathbf{a}^{+}$ and $\mathbf{a}^{-}$ that we need to know. The first is that they are Hermitian conjugates of each other. That follows from their definitions. The other property is what really gives them juice. The commutator of $\mathbf{a}^{+}$ and $\mathbf{a}^{-}$ is

$$[\mathbf{a}^{-}, \mathbf{a}^{+}] = 1.$$

This is easy to prove. First, we use the definitions to write

$$[\mathbf{a}^{-}, \mathbf{a}^{+}] = \frac{1}{2\omega\hbar}[(\mathbf{P} - i\omega\mathbf{X}), \mathbf{P} + i\omega\mathbf{X})]$$

The next step is to use the commutation relations [**X**, **X**] = 0, [**P**, **P**] = 0, and [**X**, **P**] = *iℏ*. Apply these to the above equation, and you will quickly find that